Hyperspectral Target Detection Based on Interpretable Representation Network

Dunbin Shen, Xiaorui Ma, *Member, IEEE*, Wenfeng Kong, Jianjun Liu, *Member, IEEE*, Jie Wang, *Senior Member, IEEE*, and Hongyu Wang, *Member, IEEE*

***Abstract*— Hyperspectral target detection (HTD) is an important issue in Earth observation, with applications in both military and civilian domains. However, conventional representation-based detectors are hindered by the reliance on the unknown background dictionary, the limited ability to capture nonlinear representations using the linear mixture model (LMM), and the insufficient background-target recognition based on handcrafted priors. To address these problems, this article proposes an interpretable representation network that intuitively realizes LMM for HTD, making nonlinear feature expression and physical interpretability compatible. Specifically, a subspace representation network is designed to separate the background and target components, where the background subspace can be adaptively learned. In addition, to further enhance the nonlinear representation and more accurately learn the coefficients, a lightweight multiscale Transformer is proposed by modeling long-distance feature dependencies between channels. Furthermore, to supplement the depiction for target–background discrimination, a constrained energy minimization (CEM) loss is tailored by minimizing the output background energy and maximizing the target response. The effectiveness of the proposed method is demonstrated on four benchmark datasets, showing its superiority over state-of-the-art methods. The code for this work is available at https://github.com/shendb2022/HTD-IRN for reproducibility purposes.**

***Index Terms*— Constrained energy minimization (CEM) loss, deep learning, hyperspectral target detection (HTD), physical interpretability, subspace representation.**

I. Introduction

HYPERSPECTRAL images (HSIs) comprise tens to hundreds of narrow spectral bands ranging from visible to infrared, thereby providing more sufficient spectral information for characterizing the materials of ground objects than panchromatic images and multispectral images [1], [2]. The high-spectral-resolution information inspires many tasks in Earth observation, such as classification, detection, and tracking. Unlike traditional box-level detection that focuses on spatial information, hyperspectral target detection (HTD) aims to separate pixel-level or subpixel-level targets from vast surrounding background interference based on subtle spectral feature discrepancies, enabling detection under poor conditions, such as low spatial resolution and occlusion. The unique advantage of HTD has boosted many practical applications with promising performance, such as military reconnaissance [3], mineral exploration [4], environmental monitoring [5], and precision agriculture [6].

HTD can be classified into two main types based on the availability of target information: supervised detection and anomaly detection. In supervised detection, the primary objective is to identify known targets under the guidance of representative target spectra. On the other hand, anomaly detection does not rely on specific target knowledge but instead leverages the spatial–spectral differences between anomalies and their surrounding backgrounds to detect objects. Common anomaly detection approaches include statistical distribution-based methods [7], [8], [9], representation-based (R-B) methods [10], [11], and feature learning-based methods [12], [13]. Anomaly detection is particularly suitable for practical scenarios, such as rescue operations, geological surveillance, and safety control. However, supervised detection can often achieve higher accuracy compared with anomaly detection due to the presence of target spectra. Despite potential inaccuracies in target spectra, weakly supervised detection has also gained attention in recent years. For instance, multiple-instance learning has been applied to achieve satisfactory performance in HTD, even when dealing with noisy and inaccurate target labels [14], [15]. This article focuses on supervised detection using a single target spectrum, which is commonly employed in the field.

In the past few decades, various detection methods have been developed. Early classical detectors originated from signal processing, and some methods require the background to obey a certain distribution, such as the multivariate Gaussian distribution, which is unrealistic in practice [16], [17]. In contrast, R-B detectors, not limited by ideal assumptions, transform the detection task into an optimization problem by decomposing the image into background and target components and depicting them using corresponding handcrafted priors. With the boom in spectral unmixing, sparse representation [18] and low-rank representation [19] have been

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introduced into HTD, leading to significant improvements in detection accuracy. Since the mixture property of HSI pixels can be modeled based on linear mixture models (LMMs), such methods are physically interpretable and capable of subpixel detection. However, R-B detectors still face several challenges. First, the handcrafted priors may not sufficiently depict target–background discrimination. Popular priors such as sparsity [20], low rank [21], smoothness [22], and nonlocal [23] can model the common properties of HSI but may struggle to mine underlying and intrinsic correlations to identify background and target. Second, the background dictionary is unknown and often needs to be constructed in advance. Although many background estimation methods have been proposed and proven to be effective, such as orthogonal projection [24], cluster screening [25], [26], and optimization learning [27], the asynchronous estimation between the dictionary (or subspace) and coefficients cannot guarantee global optimality, thereby affecting the final detection performance. Finally, it is hard for R-B methods to extract nonlinear features of HSI. Some researchers have introduced kernel methods to make nonlinear data linearly separable by mapping into high-dimensional space [28], [29]. Shi et al. [30] adopted the bilinear mixing model to capture the background-target interaction according to the multi-instance scattering. Nonetheless, further research is required to address the challenge of nonlinear representation.

Due to the strong ability to extract nonlinear features and learn underlying distributions, the application of deep learning to HTD has recently attracted attention. However, it faces two challenging issues, namely, insufficient labeled samples and unbalanced classes, where the target samples present a very low proportion. To address this difficulty, one feasible solution is to expand training samples by data augmentation and perform binary classification based on sufficient target and background pixels. For example, Zhang et al. [31] extended the training samples by generating potential target signatures using an autoencoder and selecting background samples based on linear prediction. They then constructed "target–target" and "target–background" pixel pairs to train similarity discrimination using convolutional neural network. An alternative approach is to utilize unsupervised learning to extract informative features that aid in distinguishing between background and target samples. For instance, Shi et al. [32] proposed a deep spatial–spectral network in an unsupervised fashion to extract features of interest, with the original HSI and a corresponding region-of-interest map as inputs. The detection map was then obtained by calculating the Euclidean distance between extracted features and the target spectrum as detection values. Due to their inherent capability to automatically learn features and patterns, current deep learning-based (DL-B) detectors have achieved moderate levels of success. However, these deep models are typically data-driven and can be likened to hierarchical "black boxes", which often lack robust physical interpretability. The mixture property of HSI pixels is usually neglected, and the common properties such as sparsity and low rank are rarely considered, which limits the detection performance.

To address the aforementioned issues, we propose an HTD approach based on an interpretable representation network (HTD-IRN) that considers both nonlinear feature representation and physical interpretability. Specifically, we convert the subspace representation model (SRM) for HTD into an IRN called the subspace representation network (SRN). The SRN can adaptively separate the target and background components, and unmix them into subspaces (endmembers) and coefficients (abundances). The background subspace and coefficients are jointly estimated, resolving the issue of asynchronous estimation. To further improve the nonlinear feature representation and more accurately learn the encoding coefficients, we introduce a lightweight multiscale Transformer module that captures the contextual relationship between channels in different scales and significantly reduces computational complexity compared to the original vision Transformer (ViT) [33]. Moreover, we propose a constrained energy minimization (CEM) loss to more accurately separate the background and targets based on handcrafted priors. The CEM loss minimizes the output energy of background pixels based on the closeness to the target spectrum while maximizing the response of target pixels to 1, thereby suppressing background information and highlighting target information as much as possible.

The novelty and contributions of our work can be summarized as follows.

1) We propose a deep SRN to adaptively separate the target and background components and unmix them into subspaces (endmembers) and coefficients (abundances), which combines nonlinear feature representation with physical interpretability.
2) We propose a lightweight multiscale Transformer module to strengthen the nonlinear feature extraction and more accurately learn the encoding coefficients, which models long-distance feature dependencies between channels.
3) We propose a CEM loss to more accurately distinguish the background and targets, which highlights the targets with the maximum response and suppresses the background with discriminatively minimized output energy.

The remainder of this article is organized as follows. First, we briefly survey the related works about HTD in Section II. Second, we present the preliminary knowledge for the proposed method in Section III. Next, in Section IV, the proposed HTD-IRN method is introduced in detail. Then, the experiments and corresponding analysis are presented in Section V. Finally, the conclusion is drawn in Section VI.

II. Related Work

Numerous methods have been developed for HTD, which can be roughly divided into three categories: classical detectors, R-B detectors, and DL-B detectors.

A. Classical Detectors

Classical detectors are usually derived from signal processing, and some of them construct decision functions adhering to certain statistical assumptions like Gaussian distribution. Such methods include an adaptive coherence estimator [16], a matched subspace detector [17], CEM [34], a spectral angle mapper (SAM) [35], a spectral matched filter [36], orthogonal subspace projection [37], and so on. Among them, CEM is one of the most famous classical detectors, which assumes that the

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desired detection map is energy-minimized and the response of the target spectrum is maximum, such that the target can be highlighted and the background can be suppressed. Due to the complexity of data distribution and a lack of nonlinear representation, classical detectors usually perform weak performances in real-world datasets. To address these issues, some improved versions have emerged. The kernel-based variants have been proposed to solve nonlinear analysis problems [28], [38], [39], [40]. However, the reliance on certain assumptions of background distribution still exists, resulting in unavoidable performance degradation in real scenarios. The hierarchical CEM (hCEM) takes the CEM algorithm as the basic module and adds a nonlinear suppression function to form a multilayer detector [41]. To further improve nonlinear discriminative and generalization capabilities, ensemble-based cascaded CEM (E-CEM) introduces L2-norm regularization and incorporates ideas from ensemble learning, including multiscale scanning, stochastic averaging, and hierarchical learning [42]. Considering the joint exploitability of spectral and spatial information, a multiscale-superpixel-based CEM detector with sparse constraint was proposed in [43] to capture the textures of the image. These derivatization methods can achieve satisfactory performance to a certain extent due to the progress in extracting multilevel, multiscale, and spatial–spectral features. However, the nonlinear feature representation is still limited compared to DL-B methods. In addition, energy minimization treats all pixels equally and overlooks their correlation and differences, which can lead to a loss of target information in certain scenarios.

B. R-B Detectors

R-B detectors have received considerable attention for interpretable modeling of HTD without relying on statistical assumptions. Such methods usually transform the detection task into an optimization problem based on LMM or nonlinear LMM (NLMM) and various handcrafted priors, such as sparsity, low rank, and smoothness. For example, the sparse R-B methods [44], [45] model each hyperspectral pixel as a linear mixture of several spectral signals, and the sparse abundances are used for detection. Taking into account the competitive and cooperative effects of spectral signals, Li et al. [46] proposed a combined sparse and collaborative representation (CSCR) detector. Considering the linear correlation of background information, Bitar et al. [47] constructed a low-rank learning model. Combining the sparsity of the target coefficients and the low rank of the background coefficients, Wang et al. [48] developed a dictionary-based model and replaced the standard nuclear norm with a reweighted one that considers the contribution of each singular value. The local homogeneity of background information also indicates the smoothness of space; thus, Cheng and Wang [27] proposed a decomposition model with background dictionary learning (DM-BDL) based on background smoothness and target sparsity. In recent studies, there has been a rise in the use of tensor decomposition-based methods [49], [50], [51] to preserve the spatial–spectral structure of HSIs and capture the underlying correlations. Considering the nonlocal self-similarity and the rank-1 canonical decomposition attribute, Zhao et al. [51] proposed a three-order tensor-based detector to extract the spatial–spectral features where appropriate dictionaries can be obtained and the number of training samples can be reduced. As mentioned in the introduction, although R-B methods have strong interpretability and promising performance, they are limited by the unknown background dictionary, inaccurate depiction for target–background separation, and nonlinear representation.

C. DL-B Detectors

Interest in applying deep learning to HTD has recently increased significantly due to powerful feature representation capabilities and fast parallel computation. Considering that HTD faces the challenges of limited unlabeled samples and background-target unbalance, current DL-B detectors can be divided into two categories: classification-based detectors and unsupervised deep detectors. Classification-based detectors first generate enough training samples by data augmentation and then treat the task as a binary classification problem. For example, Li et al. [52] generated "similarity–dissimilarity" pixel pairs for training based on the reference map for classification and adopted a multilayer convolutional neural network for transferred deep learning. Considering the mixing property of hyperspectral pixels, Zhu et al. [53] combined typical background samples and known target spectrums to randomly generate target samples for target–background balance and proposed a two-stream convolutional network-based detector to extract discrimination between the targets and background. In addition, to solve the problem of similarity metric learning, Dong et al. [54] proposed a symmetric information-theoretic metric learning method, which built a neighborhood metric space to retain local discriminative information and improved the detection performance and generalization ability. On the other hand, the unsupervised deep detectors focus on extracting discriminative features to separate the background and targets. Since no additional training samples are required, self-supervised approaches are a promising option. For instance, Shi et al. [55] proposed a continuous sparse autoencoder structure with distance constraints to enhance the discriminative ability of hidden-layer features. Considering the importance of spectral consistency, Xie et al. [56] imposed spectral regularization on an autoencoder network to extract the discriminative features and then generated the detection map based on band selection and spatial filtering. Furthermore, they introduced a background learning model known as the target suppression constraint-based background learning (BLTSC) model, which exploits the spectral disparity between the background and the original HSI to facilitate detection [57]. The presence of spectral variation, where the same material can exhibit different spectra or different materials can have the same spectrum, greatly hampers the performance of detection methods. To address this issue, Li et al. [26] proposed a hierarchical denoising autoencoder network and applied a two-step subspace projection to enhance the target and suppress the background, resulting in an excellent detector. While both classes of DL-B methods can effectively extract nonlinear features, they also encounter specific challenges within their respective approaches. On the one hand, classification-based methods suffer from inaccuracies in identifying similar or dissimilar pixel pairs due to the difficulty

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in obtaining relatively pure background and target pixels. On the other hand, unsupervised deep methods typically disregard the mixed nature of pixels and fail to incorporate common prior constraints, resulting in a lack of physical interpretability and a limitation in detection performance.

Our method falls under the category of unsupervised deep detectors as it does not require labeled samples apart from the target spectrum. In contrast to typical unsupervised deep detectors that primarily focus on extracting discriminative features through self-supervision, our method introduces an IRN that considers the mixed property of pixels and the sparsity prior. First, a deep SRN is designed to adaptively separate the target and background components and unmix them into subspaces (endmembers) and coefficients (abundances), which combines nonlinear feature representation with physical interpretability. Besides, to enhance nonlinear feature representation and learn the coefficients, we propose a lightweight multiscale Transformer module by capturing long-range interdependencies between channels. Moreover, we design a CEM loss to achieve accurate separation of background and targets by minimizing the output background energy while maximizing the target response. It is worth noting that this loss directly correlates with the input and detection results, which makes our method a fully end-to-end detector, facilitating its practical application.

III. Preliminary

In this article, the HSI and the priori target spectrum are denoted by $\mathbf{X} \in \mathbb{R}^{L\times N}$ and $\mathbf{t} \in \mathbb{R}^{L\times 1}$, where L and N are the numbers of bands and pixels, respectively. The tensor form of $\mathbf{X}$ is denoted by $\mathcal{X} \in \mathbb{R}^{H\times W\times L}$, where H and W are the height and width, respectively, i.e., $N = H \times W$.

A. SRM for HTD

R-B methods have shown significant and interpretable performance in HTD tasks. Among them, the subspace representation, i.e., decomposing the image matrix into two low-dimensional submatrices, can effectively alleviate the information redundancies and reduce the computational complexity. Here, we introduce the optimization model based on the subspace representation of the image matrix for HTD.

To facilitate the separation of the background and targets, the original HSI can be decomposed as

$$\mathbf{X} = \mathbf{B} + \mathbf{T} + \mathbf{E} \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{L\times N}$, $\mathbf{T} \in \mathbb{R}^{L\times N}$, and $\mathbf{E} \in \mathbb{R}^{L\times N}$ denote the background components, the target components, and noise, respectively. To accurately and efficiently represent the background and target components, the subspace representations of $\mathbf{B}$ and $\mathbf{T}$ are introduced; thus, (1) can be reformulated as

$$\mathbf{X} = \mathbf{A}_b\mathbf{S}_b + \mathbf{t}\mathbf{S}_t + \mathbf{E} \tag{2}$$

where $\mathbf{A}_b \in \mathbb{R}^{L\times m}$, $\mathbf{S}_b \in \mathbb{R}^{m\times N}$, and $\mathbf{S}_t \in \mathbb{R}^{1\times N}$ denote the background subspace, the background coefficients, and the target coefficients, respectively. For simplicity, the prior-known target spectrum $\mathbf{t}$ usually acts as the target subspace. m is the number of spectral signals contained in $\mathbf{A}_b$, satisfying the low-dimensional property, i.e., $m + 1 \ll L$.

Based on (2), the detection problem can be translated into an optimization model of estimating the background and target coefficients by incorporating some common handcrafted priors. It is expressed as

$$\begin{aligned} &\min_{\mathbf{S}_b,\mathbf{S}_t} \frac{1}{2}\|\mathbf{X} - \mathbf{A}_b\mathbf{S}_b - \mathbf{t}\mathbf{S}_t\|_F^2 + \lambda_1\phi_1(\mathbf{S}_b) + \lambda_2\phi_2(\mathbf{S}_t) \\ &\text{s.t. } \mathbf{A}_b \geq \mathbf{0}, \quad \mathbf{S}_b \geq \mathbf{0}, \ \mathbf{S}_t \geq \mathbf{0}, \ \mathbf{1}_m^T\mathbf{S}_b + \mathbf{S}_t = \mathbf{1}_N^T \end{aligned} \tag{3}$$

where the first term is the fidelity term, the other two terms $\phi_1(\mathbf{S}_b)$ and $\phi_2(\mathbf{S}_t)$ are the prior terms imposed on the background and target coefficients, respectively, λ_1 and λ_2 are the tradeoff parameters, and $\mathbf{1}_m$ and $\mathbf{1}_N$ denote the m- and N-size column vectors of ones. To characterize the target and background coefficients, various priors can be employed. For $\phi_1(\mathbf{S}_b)$, the sparsity prior can be used to promote sparse representation, the low rank prior can capture linear correlation, and the smoothness prior can ensure local homogeneity of the background. Similarly, for $\phi_2(\mathbf{S}_t)$, the sparsity prior can capture sparse representation, and the spatial similarity prior can describe spatial similarity features of the target. These priors can be individually applied or combined flexibly to depict specific properties of the background and target. In a physical sense, both the subspaces and coefficients of the background and target should be nonnegative, and the joint coefficients should satisfy the sum-to-one constraint.

However, for model (3), the background subspace usually needs to be known in advance, which is difficult to estimate and maintain its completeness and purity. Furthermore, handcrafted priors are insufficient to accurately depict properties separating background and targets, and nonlinear features are hard to extract. In this article, we propose a deep network to represent such models, which can preserve physical interpretability, enable nonlinear representations, adaptively learn background subspaces, and efficiently solve model problems via gradient descent.

B. CEM

CEM aims to design such a finite impulse filter that the energy of background output is minimized and the response of targets is maximized, i.e., the filter result of the target spectrum is 1. It can be formulated as the following optimization problem:

$$\min_{\mathbf{W}} \frac{1}{N}\sum_{i=1}^{N}(\mathbf{W}^T\mathbf{x}_i)(\mathbf{W}^T\mathbf{x}_i)^T \quad \text{s.t. } \mathbf{W}^T\mathbf{t} = 1 \tag{4}$$

where $\mathbf{x}_i$ denotes the ith pixel and $\mathbf{W} \in \mathbb{R}^{L\times 1}$ represents the filter matrix. By using the Lagrange multiplier method, it can be reformulated as

$$\min_{\mathbf{W}} \mathbf{W}^T\mathbf{\Phi}\mathbf{W} + \mu(\mathbf{W}^T\mathbf{t} - 1) \tag{5}$$

where μ denotes a Lagrange multiplier and $\mathbf{\Phi} = (1/N)\sum_{i=1}^{N}\mathbf{x}_i\mathbf{x}_i^T$ represents the self-correlation matrix. Making the derivatives of μ and $\mathbf{W}$ 0, respectively, the solutions are

$$\mu = \frac{-2}{\mathbf{t}^T\mathbf{\Phi}^{-1}\mathbf{t}}$$

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$$\mathbf{W} = \frac{\mathbf{\Phi}^{-1}\mathbf{t}}{\mathbf{t}^T\mathbf{\Phi}^{-1}\mathbf{t}}. \tag{6}$$

Thus, the detection result of CEM is $\mathbf{W}^T\mathbf{X}$, which can be reshaped as the detection map. Motivated by the idea of CEM, we propose a CEM loss to highlight the target information and suppress the background information.

IV. Proposed Method

In this section, the proposed method is introduced in detail. We first present the overall architecture. Then, detailed descriptions of the three parts, including the deep SRN, the detection block, and the loss function, are given, respectively.

A. Overall Architecture

The overall architecture of the proposed HTD-IRN is shown in Fig. 1. It consists of three lines: the first line represents the nonnegative joint subspace formed by the concatenation of the target spectrum and the adaptively learnable background subspace; the second line represents the subspace representation of the input HSI (i.e., the deep SRN), including the encoder for extracting coefficients and the decoder for recovering the input; and the last line is the detection block, including target component synthesis and detection map generation. To better understand the proposed detector, we present the data flow here. First, the input HSI goes through a convolutional layer to extract the low-level features. After this, the low-level features pass a U-shape Transformer (Uformer) to enhance feature representation and model long-range feature interactions. Then, the enhanced features are projected into the joint coefficient space through a convolutional layer and followed by a softmax function to achieve the nonnegative and sum-to-one constraints. Finally, the joint coefficients are decoded to recover the input HSI by multiplying the joint subspaces, which are the concatenation of the background subspace and the target spectrum. The background subspace is defined as a learnable variable and followed by a rectified linear unit (ReLU) [58] activation to obtain the nonnegative constraint. In parallel with the deep SRN, the target coefficients are separated from the joint coefficients and synthesize the target components by multiplying the target spectrum. Then, the target components go through a multiscale convolutional layer to generate the detection map. Sections IV-B–IV-D, delineate the two primary constituents: the deep SRN using Uformer and the detection block.

B. Deep SRN

We represent model (3) as the deep SRN by converting the fidelity term into an encoder–decoder structure and keeping the prior terms in the loss function. Thus, the nonlinear representation and physical interpretability can be compatible, and the optimization model can be solved effectively and easily via gradient descent.

1) Extracting Coefficients With Encoder: The encoder is to learn a nonlinear mapping between the HSI and the joint coefficients as

$$\left[\mathbf{S}_b^T, \mathbf{S}_t^T\right]^T = f_{\text{en}}(\mathcal{X}, \Theta) \tag{7}$$

where f_{en} denotes the encoder estimating the background and target coefficients from the HSI, Θ denotes the parameters involved in the encoder, and $[\cdot]$ is the concatenation operation. Specifically, we first apply a 1×1 convolution to extract the low-level features and use a leaky ReLU activation to improve the nonlinearity

$$\mathcal{Y}_0 = \sigma_1\left(W_1^0\mathcal{X}\right) \tag{8}$$

where $\mathcal{Y}_0 \in \mathbb{R}^{H\times W\times C}$ is the low-level features with C channels, W_k^i represents the $k \times k$ convolution numbered i, and $\sigma_1(\cdot)$ is the leaky ReLU [59] function. Then, we design a lightweight multiscale Transformer module, i.e., the Uformer, for multiscale analysis and feature enhancement

$$\mathcal{Y}_1 = \text{Uformer}(\mathcal{Y}_0) \tag{9}$$

where $\mathcal{Y}_1 \in \mathbb{R}^{H\times W\times C}$ is the enhanced features. Finally, we project the enhanced features into the coefficient space by applying a convolution and a softmax function to obtain the desired joint coefficients as

$$\left[\mathbf{S}_b^T, \mathbf{S}_t^T\right]^T = \text{reshape}\left(\text{softmax}\left(W_1^1\mathcal{Y}_1\right)\right). \tag{10}$$

Note that the softmax function is applied along the channel dimension to ensure the nonnegative and sum-to-one constraints on the joint coefficients, and the reshape operation here is to unfold the tensor into matrix form.

2) Recovering Image With Decoder: The decoder is to recover the HSI from the joint coefficients by multiplying the joint subspaces, which can be expressed as

$$\overline{\mathcal{X}} = \text{reshape}\left([\sigma_2(\mathbf{A}_b), \mathbf{t}]\left[\mathbf{S}_b^T, \mathbf{S}_t^T\right]^T\right) \tag{11}$$

which is equivalent to

$$\overline{\mathcal{X}} = \text{reshape}(\sigma_2(\mathbf{A}_b)\mathbf{S}_b + \mathbf{t}\mathbf{S}_t) \tag{12}$$

where $\mathbf{A}_b$ is an adaptively learnable variable, the ReLU function $\sigma_2(\cdot)$ is applied to ensure the nonnegative property of background subspace, and the reshape operation here is to fold the matrix into tensor form.

C. Uformer

Inspired by the well-known U-Net [60] and Transformers [61], [62], [63], we propose a multiscale Transformer structure, i.e., Uformer, which can model long-range feature dependencies, exploit multiscale features, and reduce the computational requirements compared with the single-scale network. Thus, the enhanced features can be obtained, and the desired coefficients can be more accurately extracted.

1) Overall Structure: Fig. 2 shows the structure of the proposed Uformer. It adopts a U-shape hierarchical design and consists of three levels of feature encoding under pyramid spatial resolution. Each level contains a Transformer encoder to capture the contextual relationship. From the first level to the last level, the spatial resolution is progressively reduced by applying the average pooling operation. Inversely, the high-resolution representation is gradually recovered from the low resolution by applying transposed convolutions. It is worth noting that, when performing downsampling or upsampling operations, the channel size remains unchanged to reduce

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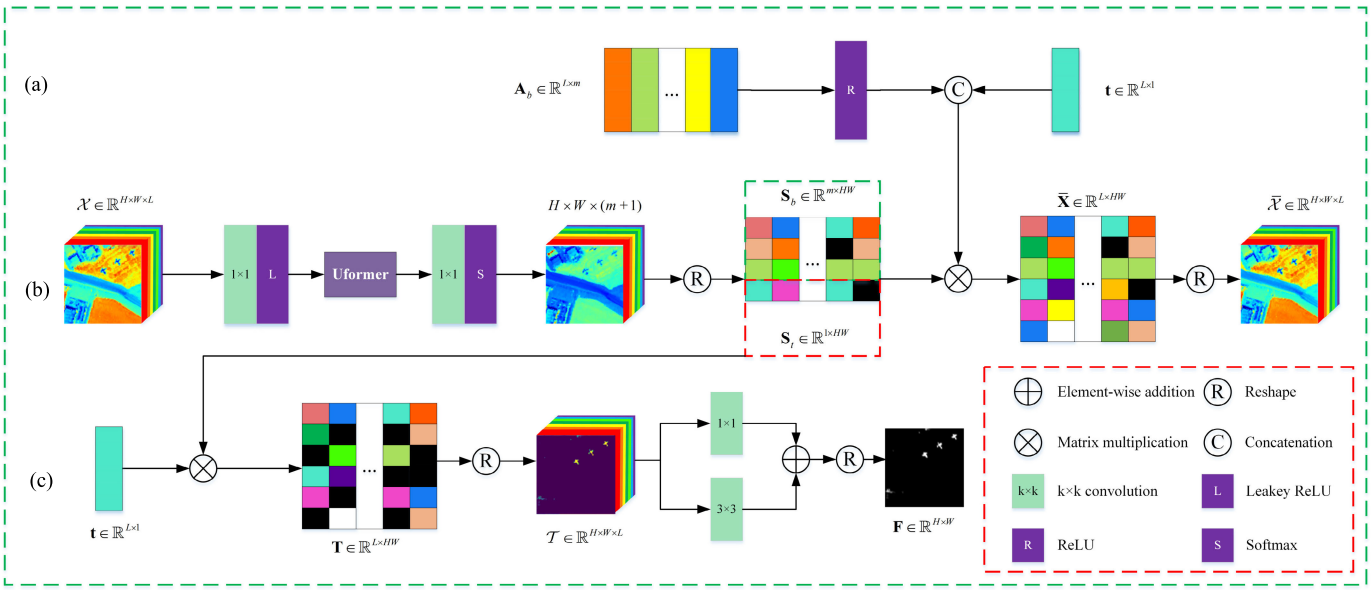


Fig. 1. Overall architecture of the proposed HTD-IRN. $\mathcal{X}$ and $\mathbf{t}$ are the inputs to the network, while $\mathbf{A}_b$ is an adaptively learnable variable. (a) Joint subspace by concatenating nonnegative $\mathbf{A}_b$ and $\mathbf{t}$. (b) Subspace representation of the input HSI, i.e., the deep SRN. (c) Detection block.

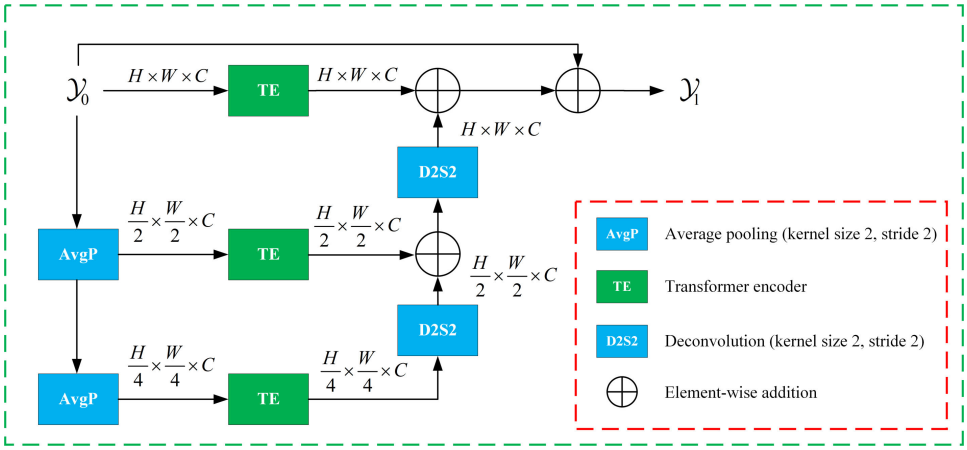


Fig. 2. Structure of Uformer. Our Ufomer is in a U-shape hierarchical design for multiscale analysis embedding with lightweight Transformer encoders.

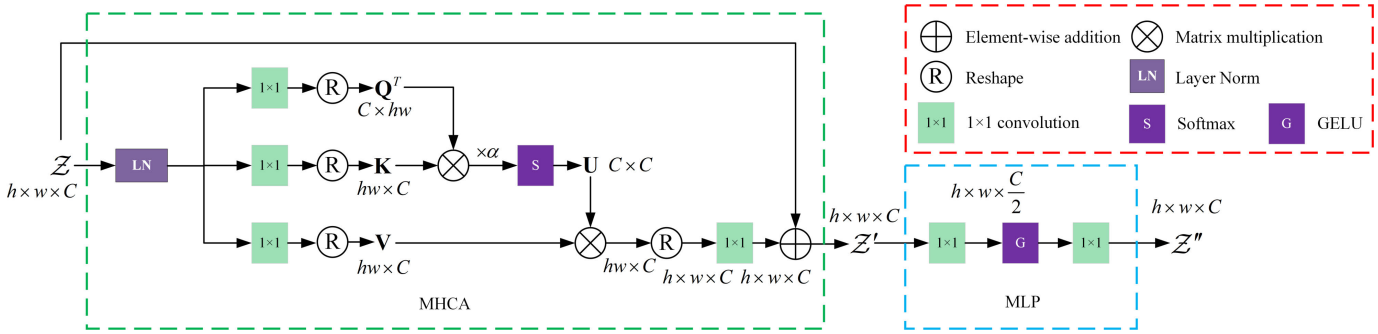


Fig. 3. Transformer encoder with multihead self-channel attention. The encoder consists of two core modules: MHCA captures long-range channel interactions, and MLP can further transform the attended features nonlinearly.

computational requirements and facilitate subsequent residual connections. To mitigate information loss during the recovery process, we integrate the coding features from the Transformer encoder with the features reconstructed from the lower level through elementwise addition. Compared with the more commonly used skip connections, it helps to maintain the lightweight structure of the proposed Uformer. The output of the first level is added to the input to obtain the enhanced features via the global residual connection, which can further preserve the main information and ensure that what Uformer learns is detailed information.

2) Transformer Encoder: The self-attention mechanism is the core idea of Transformer. However, in the classical self-attention, the computational complexity is quadratic to the spatial resolution of the input, which is not feasible for the target detection based on the whole image as input in most cases. Therefore, we adopt a lightweight variety with linear complexity to address this issue. Specifically, we model the contextual

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relationship between channels instead of the original patches or pixels. In other words, the cross-covariance between channels is computed to obtain the attention map. Besides the lightweight consideration, this design can preserve spatial information so that each pixel can be independently represented by the joint subspace. Otherwise, the spatial–spectral confusion may lead to underqualified coefficients and, finally, limit the detection performance. Fig. 3 depicts the pipeline of the proposed Transformer encoder with multihead self-channel attention (MHCA), a detailed description of which is presented in the following.

Given a tensor $\mathcal{Z} \in \mathbb{R}^{h\times w\times C}$, after going through a layer normalization, we first compute the three linear projections, i.e., Query (Q), Key (K), and Value (V) by applying 1×1 convolutions: $\mathbf{Q} = \text{reshape}(W_1^2\text{LN}(\mathcal{Z}))$, $\mathbf{K} = \text{reshape}(W_1^3\text{LN}(\mathcal{Z}))$, and $\mathbf{V} = \text{reshape}(W_1^4\text{LN}(\mathcal{Z}))$, where LN denotes the layer normalization operation and the sizes of the three projections are all $hw \times C$. Next, we generate the attention map $\mathbf{U} \in \mathbb{R}^{C\times C}$ by computing the transposed similarity between $\mathbf{Q}$ and $\mathbf{K}$, and following a softmax function. Then, the self-attention is obtained by multiplying $\mathbf{V}$. This process is expressed as

$$\begin{aligned}\mathbf{U} &= \text{softmax}(\mathbf{Q}^T\mathbf{K} \times \alpha)\\ \text{Attention}(\mathcal{Z}) &= \text{reshape}(\mathbf{VU})\end{aligned} \tag{13}$$

where α is a learnable scaling factor to adjust the magnitude of the dot product and ensure the stability of the softmax output. To further capture the relationship between different channels, similar to the classical Transformers, the idea of multiple heads is applied. Specifically, we divide the number of channels into several heads and compute the individual self-attention in parallel. For this, the three projections $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are reshaped into new projections $\mathbf{Q}' \in \mathbb{R}^{h_n\times hw\times C/h_n}$, $\mathbf{K}' \in \mathbb{R}^{h_n\times hw\times C/h_n}$, and $\mathbf{V}' \in \mathbb{R}^{h_n\times hw\times C/h_n}$, where h_n is the number of heads. Thus, the multihead self-attention is calculated by

$$\begin{aligned}\mathbf{U}' &= \text{softmax}(\mathbf{Q}'^T\mathbf{K}' \times \alpha')\\ \text{Attention}(\mathcal{Z}) &= \text{reshape}(\mathbf{V}'\mathbf{U}')\end{aligned} \tag{14}$$

where $\mathbf{U}' \in \mathbb{R}^{h_n\times C/h_n\times C/h_n}$ is the multihead attention map and α' is a learnable vector of size h_n that controls the magnitude of dot product in each head. Finally, a linear transform, i.e., a 1×1 convolution, is applied to the self-attention to obtain the residual result to which the input of the Transformer encoder is added to generate the features of interest $\mathcal{Z}' \in \mathbb{R}^{h\times z\times C}$

$$\mathcal{Z}' = W_1^5\ \text{Attention}(\mathcal{Z}) + \mathcal{Z}. \tag{15}$$

To transform the features of interest nonlinearly, the output of MHCA further goes through a multilayered perceptron (MLP) module and generates the final enhanced features $\mathcal{Z}'' \in \mathbb{R}^{h\times z\times C}$

$$\mathcal{Z}'' = W_1^7\sigma_3\left(W_1^6\mathcal{Z}'\right) \tag{16}$$

where $\sigma_3(\cdot)$ denotes the Gaussian error linear unit (GELU) [64] activation. The adopted MLP module consists of two 1×1 convolutions and a nonlinearity in the hidden layer. The effectiveness of Uformer is investigated in Section V-G.

D. Detection Block

With the learned target coefficients, the desired target components $\mathcal{T} \in \mathbb{R}^{H\times W\times L}$ can be synthesized by

$$\mathcal{T} = \text{reshape}(\mathbf{t}\mathbf{S}_t). \tag{17}$$

Since our deep SRN exploits the mixture property of hyperspectral pixels, the synthesized target components may contain subpixel targets, which demonstrates the capability of our detector in handling subpixel target detection with high accuracy.

To transform the target components into the final detection map, some works adopt the band L2-norm [27], the maximum value method [26], the nonlinear suppression fucntion [55], and so on. However, these strategies are all static and never consider spatial relationships. To address this, we propose a multiscale convolutional block to build dynamic mapping that can effectively exploit the spatial correlations in the neighborhood

$$\mathbf{F} = \text{reshape}\left(W_1^8\mathcal{T} + W_3^0\mathcal{T}\right) \tag{18}$$

where the final detection map $\mathbf{F} \in \mathbb{R}^{H\times W}$ is generated by combining the output of a 1×1 convolutional layer and a 3×3 convolutional layer through elementwise addition, as illustrated in Fig. 1(c). This fusion strategy enables the effective exploitation of both pixel-level spectral information and block-level spatial information to determine the presence of targets at the pixel or subpixel level. To achieve this, we adopt a multiscale analysis approach. The effectiveness of this multiscale mapping (MSM) is investigated in Section V-G.

E. Loss Function

As mentioned above, we build the network based on the fidelity term of model (3). To train the network, we transform both the fidelity and prior terms into the corresponding items of the loss function. Specifically, the fidelity term becomes the reconstruction loss, and the uncertain prior terms are specific to the sparse loss constrained on the joint coefficients for simplicity and effectiveness. Besides, the CEM loss is considered a core item to accurately separate the background and targets. Therefore, a composite loss function is tailored as

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \eta_1\mathcal{L}_{\text{spar}} + \eta_2\mathcal{L}_{\text{CEM}} \tag{19}$$

where $\mathcal{L}_{\text{recon}}$, $\mathcal{L}_{\text{spar}}$, and $\mathcal{L}_{\text{CEM}}$ denote the reconstruction loss, the sparse loss, and the CEM loss, measuring the reconstruction effect, sparsity of joint coefficients, and background-target discrimination, respectively. η_1 and η_2 are two tradeoff parameters to balance the three terms.

The reconstruction loss is to measure the difference between the input and the output of the proposed SRN. Considering that L1-norm can better preserve the texture and edge information compared with the Frobenius norm, we adopt L1-norm to perform $\mathcal{L}_{\text{recon}}$ as

$$\mathcal{L}_{\text{recon}} = \frac{1}{LHW}\|\mathcal{X} - \overline{\mathcal{X}}\|_{1,1,1} \tag{20}$$

where $\mathcal{X}$ and $\overline{\mathcal{X}}$ are the input HSI and the reconstructed HSI, respectively, and $\|\cdot\|_{1,1,1}$ denotes the L1-norm of tensor, i.e., the absolute sum of all the tensor elements.

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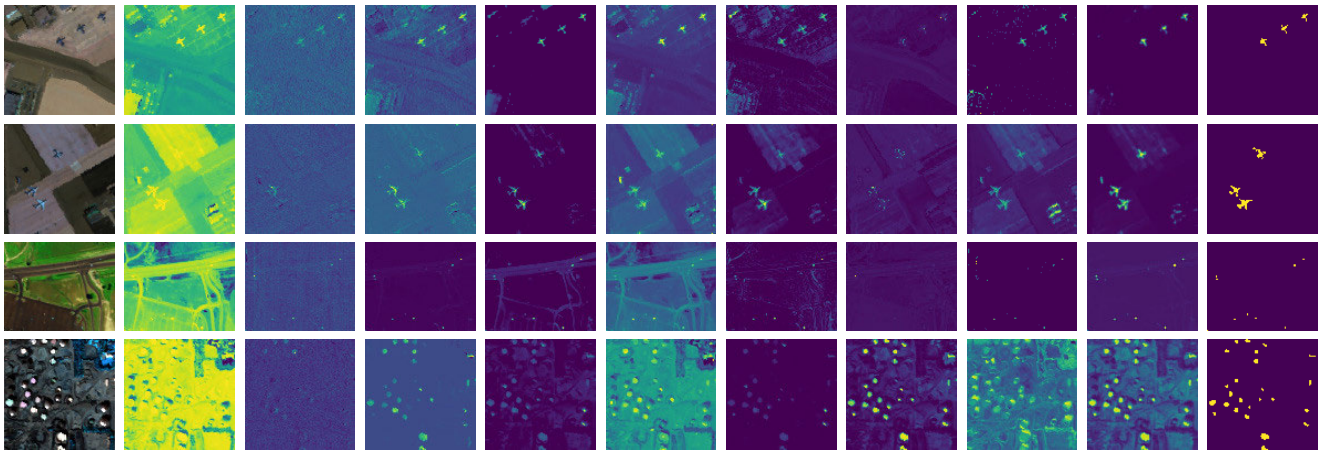


Fig. 4. Experimental datasets and detection maps of competing methods. (a) False-color image. (b) SAM. (c) CEM. (d) hCEM. (e) E-CEM. (f) CSCR. (g) DM-BDL. (h) HTD-Net. (i) BLTSC. (j) HTD-IRN. (k) Ground truth. From (top) to (bottom): San Diego I, San Diego II, HYDICE, and Urban.

The sparse loss is applied to express the sparsity of coefficients that only a few bases are responsible for representing each pixel. The sparse constraint can make the model more concise and enhance its interpretability. In this article, we adopt the classical L1-norm to constrain the sparsity of the joint coefficients as

$$\mathcal{L}_{\text{spar}} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{S}(:, i)\|_1 \tag{21}$$

where $\mathbf{S} = [\mathbf{S}_b^T, \mathbf{S}_t^T]^T$, and $\|\cdot\|_1$ denotes the L1-norm of vectors, i.e., the absolute sum of all the vector elements.

Inspired by the core idea of CEM that the target signatures are matched by maximizing the response of the known target spectrum and limiting the output energy of the background, we design a CEM loss term to highlight the target and suppress the background. Similar to model (4), we define our deep detector as a mapping $f : \mathbb{R}^L \to \mathbb{R}^1$ that can be approximated as the finite impulse filter $\mathbf{W}$. Thus, the CEM loss can be expressed as

$$\mathcal{L}_{\text{CEM}} = \frac{\eta_3}{HW}\|f(\mathcal{X})\|_F^2 + |f(\mathbf{t}) - 1| \tag{22}$$

where $f(\mathcal{X}) \in \mathbb{R}^{H\times W}$ denotes the detection map, $f(\mathbf{t}) \in \mathbb{R}$ denotes the detection response of the target spectrum, the absolute value operation $|\cdot|$ is adopted to ensure the distance positive, and η_3 is a tradeoff parameter to balance the two terms. The first term corresponds to the optimization term of CEM to minimize the energy of the filtering result, and the second term corresponds to the condition of CEM, that is, the response of the target signature is maximized by forcing the filtering result of the target spectrum to 1. As a result, the target information can be highlighted, and the background information can be suppressed. However, this energy-minimized term treats all pixels equally, which limits detection performance. To address this issue, we improve this term by introducing a weight matrix that considers the difference between each pixel and the target spectrum. Besides, since our deep detector is based on the tensor input, we expand the target spectrum to form a tensor cube for facilitating unified calculation. Therefore, the CEM loss becomes

$$\mathcal{L}_{\text{CEM}} = \frac{\eta_3}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{D}(i, j)\mathbf{F}^2(i, j) + \frac{1}{HW}\|\mathbf{G} - \mathbf{1}_{H\times W}\|_{1,1} \tag{23}$$

where $\mathbf{1}_{H\times W} \in \mathbb{R}^{H\times W}$ is a matrix of ones, $\|\cdot\|_{1,1}$ represents the L1-norm of a matrix, i.e., the absolute sum of all matrix elements, and

$$\begin{aligned} \mathbf{D}(i, j) &= \|\mathcal{X}(i, j) - \mathbf{t}\|_2^2 \\ \mathbf{F} &= f(\mathcal{X}) \\ \mathbf{G} &= f(\mathbf{t}_{H\times W\times L}) \end{aligned} \tag{24}$$

where $\|\cdot\|_2$ denotes the L2-norm of vectors, and $\mathbf{t}_{H\times W\times L} \in \mathbb{R}^{H\times W\times L}$ is a tensor cube expanded by $\mathbf{t}$. Since the weight matrix $\mathbf{D}$ measures the difference between each pixel and the target spectrum, the CEM loss can express the affinity between all pixels and the target spectrum. In other words, the closer the pixel is to the target spectrum, the less energy it weakens, and vice versa, which can avoid the loss of a large amount of target information. Therefore, the target information can be preserved, and the background information can be suppressed more effectively. The effectiveness of the regularization loss terms is verified in Section V-G.

V. Experiments

This section presents the experimental setup and corresponding discussions, including dataset description, evaluation metrics, implementation details, parameter selection, comparison methods, experimental results, and ablation studies that are carried out to evaluate the proposed approach.

A. Dataset Description

In this article, experiments are conducted on four benchmark datasets to verify the effectiveness of the proposed HTD-IRN method: San Diego I, San Diego II, HYDICE, and Urban. Fig. 4(a) and (k) displays the false-color images and ground-truth maps of the datasets, respectively. Since the target spectrum should be prior-known, in our experiments, the pixel

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with the smallest average distance from other target pixels is selected as the target spectrum.

1) The San Diego I and II datasets were collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the San Diego Airport, San Diego, California, USA. Both the two subscenes consist of 100 × 100 pixels with a spatial resolution of 3.5 m. After removing the low signal-to-noise ratio (SNR) and water vapor absorption bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), there left 189 bands in a range of 400–2500 nm and with a spectral resolution of 10 nm. Three airplanes, including 58 pixels in I and 134 pixels in II, are the targets to be detected.
2) The HYDICE dataset covers an urban area in California, and it was collected by the Hyperspectral Digital Image Collection Experiment (HYDICE) airborne sensor. It has a spatial resolution of 1 m and a spectral resolution of 10 nm. The number of bands is 162 after eliminating the water vapor absorption and noisy bands (1–4, 76, 87, 101–111, 136–153, and 198–210). The whole dataset consists of 307 × 307 pixels. To facilitate detecting anomalous targets in the upper right corner of the scene, a considered subimage of size 100 × 100 (row: 1–80, column: 188–287) is cropped for the experiments. The vehicles and roofs embedded in the scene are the anomalous targets, which include 19 pixels.
3) The Urban dataset is one of the Airport-Beach-Urban images, which was collected by the AVIRIS sensor and available online.[1] The detailed production process has been introduced in [65]. The scene contains 100 × 100 pixels with a spatial resolution of 7.1 m and 205 bands with a spectral resolution of 10 nm. Among them, the storage tanks composed of 272 pixels are the targets to be detected.

B. Evaluation Metrics

To comprehensively evaluate the detection performance, the visualized detection map, the 3-D receiver operating characteristic (3-D ROC) analysis [66], and the target–background separation diagram are adopted.

The 3-D ROC is a function of threshold τ, detection probability P_d, and false alarm rate P_f. With a given τ, P_d and P_f can be calculated by

$$P_d = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad P_f = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{25}$$

where TP and FN denote the number of target pixels with detection values greater and less than τ, respectively, while FP and TN correspond to such background pixels, respectively. Three derived 2-D ROC curves of (P_f, P_d), (τ, P_f), and (τ, P_d) are utilized to measure the detection effectiveness, background suppression, and target detectability, respectively. For the desired detector, the ideal 2-D ROC curves of (P_f, P_d), (τ, P_f), and (τ, P_d) should be close to the top left, bottom left, and top right corners of the coordinate axis, respectively.

To perform a quantitative evaluation, we calculate the area under the curve (AUC) of the three 2-D ROCs, denoted as $\mathrm{AUC}_{(P_f,P_d)}$, $\mathrm{AUC}_{(\tau,P_f)}$, and $\mathrm{AUC}_{(\tau,P_d)}$, respectively. For $\mathrm{AUC}_{(P_f,P_d)}$ and $\mathrm{AUC}_{(\tau,P_d)}$, the higher value indicates better performance, and the optimal value is 1. For $\mathrm{AUC}_{(\tau,P_f)}$, the lower value is better, and the optimal value is 0. In addition, two composite metrics of the three AUCs are also considered for all-around evaluation [67], which are expressed as

$$\mathrm{AUC}_{\mathrm{OA}} = \mathrm{AUC}_{(P_f,P_d)} + \mathrm{AUC}_{(\tau,P_d)} - \mathrm{AUC}_{(\tau,P_f)} \tag{26}$$

$$\mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}_{(\tau,P_d)}}{\mathrm{AUC}_{(\tau,P_f)}} \tag{27}$$

where the higher $\mathrm{AUC}_{\mathrm{OA}}$ and $\mathrm{AUC}_{\mathrm{SNPR}}$, the better the performance, and the optimal values are 2 and $+\infty$, respectively.

The target–background separation diagram is drawn using the Box–Whisker plot to analyze the separability between the targets and background. The longer distance and smaller overlap between the target and background boxes indicate better separation capacity.

C. Implementation Details

For the proposed network, the number of convolutional kernels C in each hidden layer is set to 128, and the number of heads h_n in Uformer is empirically set to 4. The learnable background subspace $\mathbf{A}_b$ is initialized with standard normal distribution, and the scaling factor α' in Uformer is initialized as a vector of ones. To train the network, the Adam optimizer [68] is chosen to minimize the loss. Since our method is unsupervised and is designed for target detection based on the whole image, there is no need to construct the training and test sets. Thus, for each dataset, the whole image is used as input and the batch size is 1. The learning rate is initially set to 0.001 and decays half every 1000 iterations. The total number of iterations is 5000. The regularization parameters η_2 and η_3 in the loss function are uniformly set to 1 and 0.1 for all the datasets, respectively. Considering the impact of m and η_1 on detection performance, we analyze and determine them in Section V-D. The experiments of the proposed method were all implemented in Python 3.6.13 and PyTorch 1.10.1, using an Intel[2] Core[3] i7-8700K CPU, 62-GB RAM, and an NVIDIA GeForce RTX 2080Ti 11-GB GPU server. To illustrate the convergence of the proposed approach, the training loss curves of four datasets are displayed in Fig. 5, recorded every 100 iterations. It can be seen that the training loss drops sharply in the first few epochs and then converges in a small fluctuation.

D. Parameter Selection

The proposed method has two important undetermined parameters, the background subspace dimension m and the sparsity weight η_1, which control the number of spectral bases contained in the background subspace and the sparsity of the joint coefficients, respectively.

To discuss the effect of these two parameters jointly, we investigate $\mathrm{AUC}_{(P_f,P_d)}$ as a function of two variables m and η_2. The 3-D surface drawings are shown in Fig. 6, where the peak value is marked in blue. It can be observed that

[1] http://xudongkang.weebly.com/

[2] Registered trademark.

[3] Trademarked.

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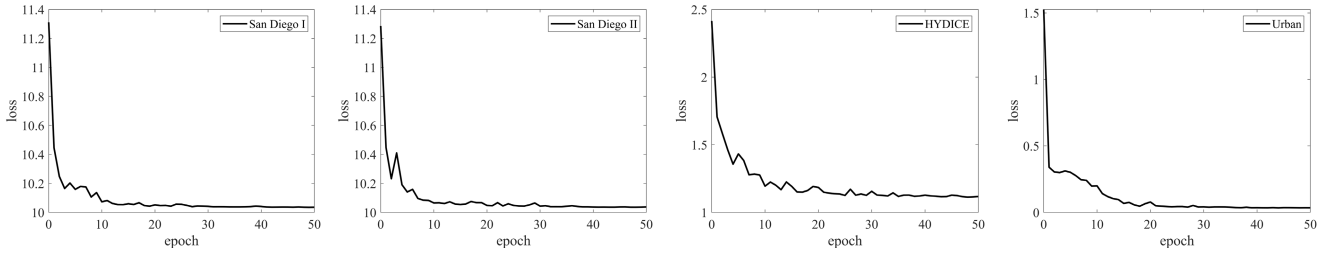


Fig. 5. Training loss of HTD-IRN over different epochs. (a) San Diego I. (b) San Diego II. (c) HYDICE. (d) Urban.

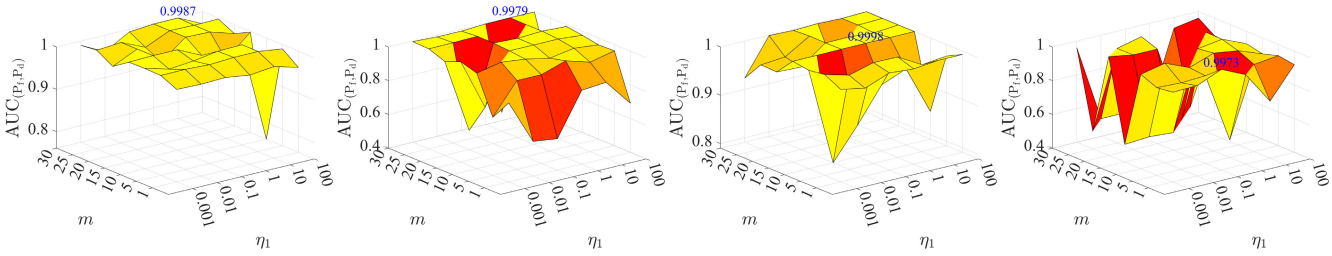


Fig. 6. Effect of parameters m and η_1 for HTD-IRN. (a) San Diego I. (b) San Diego II. (c) HYDICE. (d) Urban.

the optimal values of m and η_1 are highly dependent on the unique properties of the datasets themselves. For example, for the Urban dataset, only one spectral base is enough for representation since it has simple and homogeneous scenes. In addition, sparsity is partly related to the diversity of spectral bases. For example, the two San Diego datasets need more spectral bases to construct the background subspace, but only a few are responsible for each pixel. Therefore, their coefficients will be relatively sparser than that of the other two datasets. According to the peak values, we set (m, η_1) as (30, 10), (30, 10), (15, 1), and (1, 0.1) for the four datasets, respectively. It can also be found that the optimal values of m are much less than the number of bands, which indicates that the HSI lies in a low-dimensional subspace and can be represented by a few spectral signatures.

E. Comparison Methods

To illustrate the effectiveness and superiority, the proposed method is compared with several representative methods: SAM [35], CEM [34], hCEM[4] [41], E-CEM[5] [42], CSCR[6] [46], DM-BDL[7] [27], HTD-Net[8] [31], and BLTSC[9] [57]. All the competing methods are tuned to their ideal level for a fair comparison. Specifically, for hCEM, the two key parameters λ and ϵ are set to (200, 0.004), (200, 0.002), (1, 0.006), and (4, 0.001) for the four datasets, respectively. For E-CEM, we keep the number of layers, the number of CEMs in each layer, and multiscale scanning sizes as 10, 6, and $[(1/4)L, (1/2)L, (3/4)L, L]$ for all datasets, respectively, where L denotes the number of wavebands. The regularization parameter is set to random$(\lambda/(1+\lambda), \lambda)$, where λ is selected as 0.2, 0.1, 1.0, and 1.0 for the four datasets, respectively. For CSCR, we set the dual-window size $(w_{\text{in}}, w_{\text{out}})$ as (13, 11), (15, 13), (9, 3), and (11, 9) for San Diego I, San Diego II, HYDICE, and Urban, respectively. In addition, the regularization parameter (λ_1, λ_2) is selected as (0.1, 1.0), (0.1, 1.0), (0.1, 0.1), and (0.1, 0.1), respectively. For DM-BDL, the regularization parameter γ and the number of background atoms m are set to (10, 5), (5, 2), (25, 8), and (30, 4) for San Diego I, San Diego II, HYDICE, and Urban, respectively. In addition, two penalty parameters λ and β in the decomposition model are set to (0.001, 0.01), (0.001, 0.1), (0.1, 0.1), and (1.0, 1.0) for the four datasets, respectively. For HTD-Net, the number of augmented target and background pixels for all datasets is set to 1000 and 500, respectively. In the training process, the learning rate is initially set to 0.01, which decays by a factor of 0.1 every 80 000 iterations. The batch size is 256, and the total number of epochs is 30. For BLTSC, to select background samples, the threshold ϵ is set to 0.12, and 75% of them are randomly selected as training samples. The number of nodes and depth is set to 50 and 2, respectively. During training, the learning rate is 10^{-4}, the batch size is 64, the total number of epochs is 100, and the target suppression loss is calculated by maximizing the top 20 smallest SAM values. In the nonlinear background suppression, the parameter λ is kept as 10 for all the datasets.

F. Experimental Results

Fig. 4 presents the detection maps obtained by the seven competing methods on the four benchmark datasets. It can be found that, compared with other competitors, HTD-IRN achieves a satisfactory balance in highlighting the target and suppressing the background. For example, in San Diego II, it is difficult to detect the top plane because some dim and small targets are located in it. As can be seen in Fig. 4(b2)–(j2), SAM and CSCR exhibit a large number of false alarms, while CEM and HTD-Net demonstrate a limited ability to detect most target pixels, leading to an incomplete shape

[4]http://levir.buaa.edu.cn/Code.htm
[5]http://levir.buaa.edu.cn/Code.htm
[6]https://fdss.bit.edu.cn/pub/fsyxhyxtktz/yjdw/js/b153191.htm
[7]https://github.com/FDU-ctk/HSI-detection
[8]https://fdss.bit.edu.cn/pub/fsyxhyxtktz/yjdw/js/b153191.htm
[9]https://github.com/zhangxin-xd/BLTSC

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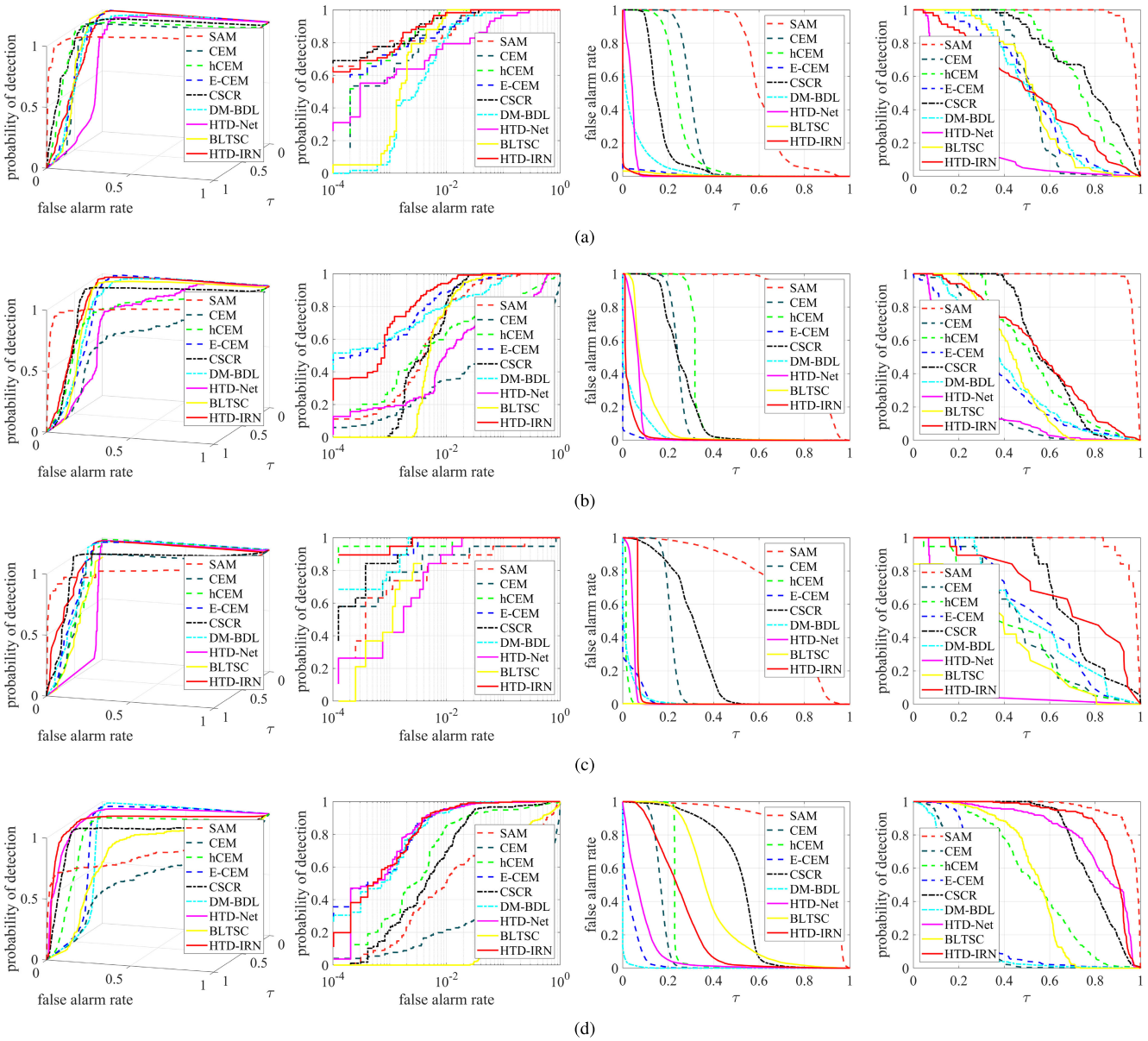


Fig. 7. ROC curves of competing methods on four datasets. (a) San Diego I. (b) San Diego II. (c) HYDICE. (d) Urban. From (left) to (right): 3-D ROC curve, 2-D ROC curve of (P_f, P_d), 2-D ROC curve of (τ, P_f), and 2-D ROC curve of (τ, P_d).

representation. Although hCEM, DM-BDL, and BLTSC perform relatively well in detecting the position of targets, their ability to suppress background interference is inferior. E-CEM achieves satisfactory performance, but our HTD-IRN shows a better capacity for target enhancement. As another example, the target pixels in Urban occupy a large proportion of the entire scene, which makes the detection uncommon. As can be seen in Fig. 4(b4)–(j4), except SAM, CEM, CSCR, and BLTSC, all other methods successfully separate the target and background. However, E-CEM and DM-BDL are weak in highlighting targets. hCEM, HTD-Net, and HTD-IRN show relatively promising performances, where hCEM suppresses the background the best, while HTD-IRN salients the target the best. The main reason for this is that HTD-IRN seamlessly integrates the SRM with deep learning, allowing it to extract distinctive nonlinear features and effectively handle subpixel detection.

The 3-D and derived 2-D ROC curves for the four datasets are shown in Fig. 7. We can find that the 2-D ROC curves of (P_f, P_d) of HTD-IRN for all four datasets are almost located in the upper left corner, which indicates its promising and robust performance. The 2-D ROC curves of (τ, P_f) and (τ, P_d) also illustrate that HTD-IRN is more reliable compared with other competing methods overall. Besides, Fig. 8 displays the target–background separation diagram. It can be seen that HTD-IRN has a better capacity for separating the background and targets, especially in Urban, where the two boxes present a large distance and small overlap.

Tables I–IV list the AUC values of the seven competing methods on the four datasets, respectively, where the best results are marked in boldface and the second-best are underlined. It is evident that HTD-IRN produces the most accurate and robust results in terms of the five metrics compared to other competitors. As for $\mathrm{AUC}_{(P_f, P_d)}$, HTD-IRN outperforms

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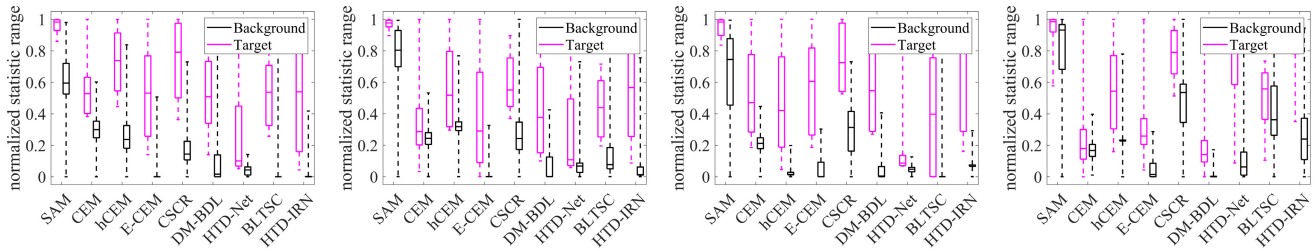


Fig. 8. Target–background separation diagram of competing methods on four datasets. (a) San Diego I. (b) San Diego II. (c) HYDICE. (d) Urban.

TABLE I

QUANTITATIVE RESULTS AND RUNNING TIME (IN SECONDS) FOR SAN DIEGO I. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Method	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}	Time
SAM	0.9966	**0.9730**	0.6188	1.3508	1.5724	**0.01**
CEM	0.9968	0.5277	0.2996	1.2249	1.7613	0.02
hCEM	0.9986	0.7421	0.2477	1.4930	2.9960	0.17
E-CEM	0.9969	0.5324	0.0093	1.5200	57.2473	9.23
CSCR	0.9985	0.7689	0.1635	**1.6039**	4.7028	20.40
DM-BDL	0.9930	0.5335	0.0481	1.4784	11.0915	5.07
HTD-Net	0.9823	0.1773	0.0396	1.1200	4.4773	30.87
BLTSC	0.9975	0.5326	0.0176	1.5125	30.2614	1.37
HTD-IRN	**0.9987**	0.5328	**0.0036**	1.5279	**148.0000**	0.26

TABLE II

QUANTITATIVE RESULTS AND RUNNING TIME (IN SECONDS) FOR SAN DIEGO II. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Method	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}	Time
SAM	0.9889	**0.9746**	0.8069	1.1566	1.2078	**0.02**
CEM	0.7080	0.3051	0.2431	0.7700	1.2550	0.03
hCEM	0.9128	0.5387	0.3203	1.1312	1.6819	0.23
E-CEM	0.9951	0.3492	**0.0038**	1.3405	**91.8947**	8.41
CSCR	0.9937	0.5865	0.2546	1.3256	2.3036	21.65
DM-BDL	0.9878	0.4050	0.0339	1.3589	11.9469	2.35
HTD-Net	0.8973	0.1793	0.0626	1.0140	2.8642	30.68
BLTSC	0.9919	0.4358	0.1012	1.3265	4.3063	1.27
HTD-IRN	**0.9979**	0.5640	0.0273	**1.5346**	20.6593	0.28

TABLE III

QUANTITATIVE RESULTS AND RUNNING TIME (IN SECONDS) FOR HYDICE. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Method	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}	Time
SAM	0.9820	**0.9668**	0.7043	1.2445	1.3727	**0.01**
CEM	0.9535	0.5186	0.2125	1.2596	2.4405	0.03
hCEM	0.9993	0.4689	**0.0175**	1.4507	26.7943	0.18
E-CEM	0.9997	0.5870	0.0212	1.5655	**27.6887**	4.45
CSCR	0.9996	0.7281	0.3024	1.4253	2.4077	14.20
DM-BDL	0.9995	0.5739	0.0213	1.5521	26.9437	5.06
HTD-Net	0.9966	0.1617	0.0459	1.1124	3.5229	22.81
BLTSC	0.9200	0.4002	0.0232	1.2970	17.2500	1.23
HTD-IRN	**0.9998**	0.7286	0.0685	**1.6599**	10.6365	0.26

TABLE IV

QUANTITATIVE RESULTS AND RUNNING TIME (IN SECONDS) FOR URBAN. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Method	$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$	AUC_{OA}	AUC_{SNPR}	Time
SAM	0.8487	**0.9624**	0.8721	0.9390	1.1035	**0.03**
CEM	0.5857	0.1995	0.1673	0.6179	1.1925	0.07
hCEM	0.9691	0.5431	0.2333	1.2789	2.3279	0.20
E-CEM	0.9968	0.2833	0.0320	1.2481	8.8531	10.83
CSCR	0.9790	0.7872	0.5016	1.2646	1.5694	19.11
DM-BDL	0.9969	0.1657	**0.0025**	1.1601	**66.2800**	4.70
HTD-Net	0.9964	0.8189	0.0797	**1.7356**	10.2748	31.32
BLTSC	0.8087	0.5350	0.3929	0.9508	1.3617	1.26
HTD-IRN	**0.9973**	0.8736	0.2447	1.6262	3.5701	0.27

other methods on the four datasets. Taking San Diego II as an example, the value of $AUC_{(P_f,P_d)}$ is 0.9979, which is 0.0090, 0.2899, 0.0851, 0.0028, 0.0042, 0.0101, 0.1006, and 0.0060 larger than those of SAM, CEM, hCEM, E-CEM, CSCR, DM-BDL, HTD-Net, and BLTSC, respectively. The average running time of the competing methods is also recorded in these four tables. It is observed that HTD-IRN takes less time than other competing methods except for the classical methods and hCEM, which demonstrates its efficiency. In agreement with the aforementioned analysis, HTD-IRN achieves the most satisfactory results on the four datasets, effectively and efficiently.

G. Ablation Studies

1) Effectiveness of Uformer: To verify the functionality and effectiveness of Uformer, ablation experiments are performed on four datasets. Fig. 9 displays the 2-D ROC curves of (P_f, P_d) and the corresponding AUC scores on the four datasets. For convenience, the proposed method is denoted as "Uformer" and "Uformer-" before and after Uformer ablation. We can find that, with Uformer, the detection performance is improved on the first three datasets and kept on Urban. This can be attributed to the contribution of Uformer, which learns more accurate and discriminative coefficients by capturing long-range contextual relations between channels. Considering that the abundance maps can intuitively reflect the quality of estimated coefficients and are widely used in unmixing approaches [69], [70], [71], Figs. 10 and 11 present the abundance maps of the background and the target, respectively. The visualized maps are obtained by calculating the L2-norm of the coefficients. It can be observed that, on the first three datasets, with Uformer, the background coefficients become

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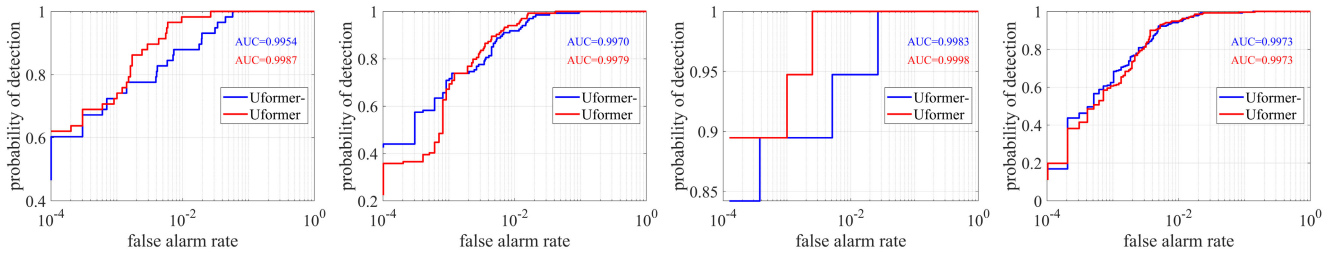


Fig. 9. 2-D ROC curve of (P_f, P_d) for Uformer ablation ("Uformer-" means not to use Uformer). (a) San Diego I. (b) San Diego II. (c) HYDICE. (d) Urban.

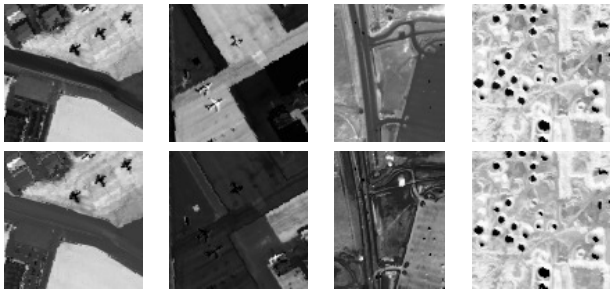


Fig. 10. Comparison of background abundances (coefficients) for Uformer ablation. (a) San Diego I. (b) San Diego II. (c) HYDICE. (d) Urban. First row: Uformer-. Second row: Uformer.

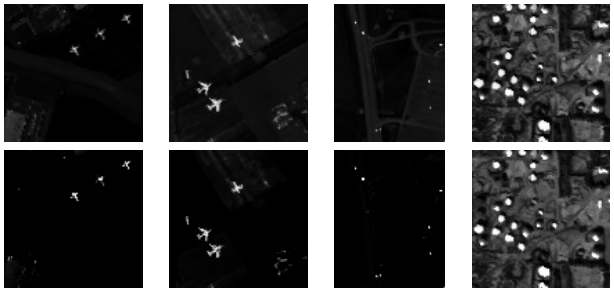


Fig. 11. Comparison of target abundances (coefficients) for Uformer ablation. (a) San Diego I. (b) San Diego II. (c) HYDICE. (d) Urban. First row: Uformer-. Second row: Uformer.

more complete and detailed, and the target coefficients become clearer and more prominent. For Urban, the coefficients remain almost unchanged, which is consistent with the final detection performance. The reason why Uformer performs poorly on Urban is that its background information is too homogeneous (the background subspace consists of one atom), so the previous convolution layer is sufficient to extract the main information, leaving little room for Uformer to capture additional detailed information, while, in real scenarios, since the background information is usually complex and varied, Uformer can extract more detailed features beyond the previous convolutional layer by capturing the contextual relationships between channels. As a result, more accurate and discriminative coefficients can be learned, leading to more promising detection performance.

To further demonstrate the superiority of Uformer, we conducted a performance comparison by replacing Uformer with five existing advanced network modules: the residual block (ResBlock) [72], U-Net [60], the squeeze-and-excitation block (SE-Block) [73], the convolutional block attention module (CBAM) [74], and ViT [33]. These modules were selected

TABLE V

PERFORMANCE COMPARISON OF UFORMER AND EXISTING ADVANCED NETWORK MODULES BY AUC$_{(P_f,P_d)}$ ON THE FOUR DATASETS. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Method	San Diego I	San Diego II	HYDICE	Urban	Average
ResBlock	0.9810	0.9503	0.9664	0.8924	0.9475
U-Net	0.9753	0.9096	0.9354	0.9665	0.9467
SE-Block	<u>0.9970</u>	**0.9981**	<u>0.9995</u>	**0.9973**	<u>0.9980</u>
CBAM	0.9958	0.9978	0.9978	<u>0.9972</u>	0.9972
ViT	0.9946	0.7589	0.8158	0.9966	0.8915
Uformer	**0.9987**	<u>0.9979</u>	**0.9998**	**0.9973**	**0.9984**

based on their powerful performance in various computer vision tasks. ResBlock extracts local spatial features and avoids information loss, U-Net excels at extracting multiscale features using an encoder–decoder structure, SE-Block focuses on learning channel weights to pay attention to channels of interest, CBAM combines both channel and spatial attention, and ViT captures nonlocal spatial independence based on multihead self-attention. The quantitative results in Table V demonstrate that the channel-based modules (SE-Block and Uformer) perform better than the spatial-based ones (ResBlock, U-Net, and ViT). The superiority of channel-based modules lies in their ability to extract discriminative features from individual pixels while preserving spatial information. In contrast, spatial-based modules may introduce spatial–spectral confusion and blurring, leading to underqualified abundances. In addition, the spatial attention in CBAM can potentially blur the feature map and alter the spatial context of abundances. Both SE-Block and Uformer can preserve morphological information, but Uformer achieves better overall performance on the four datasets due to the stronger ability of multihead self-attention to model complex channelwise interdependences compared to channel weighting. In summary, compared with some existing advanced modules, Uformer demonstrates effectiveness and superiority in extracting more accurate abundances and improving detection performance.

2) Effectiveness of Loss Function: To verify the effectiveness of the proposed composite loss function, ablation experiments are performed for each regularization term. Table VI shows the results of eight combinations, where the CEM loss $\mathcal{L}_{\text{CEM}}$ is divided into the constrained loss $\mathcal{L}_{\text{C}}$ and the energy-minimized loss $\mathcal{L}_{\text{EM}}$ corresponding to the second term and the first term in (23), respectively. It can be seen that the CEM loss is critical for the proposed method, without which the detection performance is dramatically dropped.

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TABLE VI

EFFECTIVENESS OF LOSS TERMS BY $AUC_{(P_f,P_d)}$ ON THE FOUR DATASETS. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Loss	San Diego I	San Diego II	HYDICE	Urban
$\mathcal{L}_{\text{recon}}$	0.7593	0.7062	0.9937	0.6577
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{spar}}$	0.7724	0.7063	0.9961	0.6587
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{C}}$	0.8193	0.5300	0.9567	0.9635
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{EM}}$	0.6324	0.5794	0.9897	0.8857
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{CEM}}$	<u>0.9916</u>	<u>0.9962</u>	<u>0.9995</u>	<u>0.9829</u>
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{spar}} + \mathcal{L}_{\text{C}}$	0.5347	0.7872	0.9128	0.9771
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{spar}} + \mathcal{L}_{\text{EM}}$	0.6339	0.5778	0.9908	0.8857
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{spar}} + \mathcal{L}_{\text{CEM}}$	**0.9987**	**0.9979**	**0.9998**	**0.9973**

TABLE VII

PERFORMANCE COMPARISON OF DIFFERENT MAPPING METHODS BY $AUC_{(P_f,P_d)}$ ON THE FOUR DATASETS. BOLD REPRESENTS THE BEST RESULT WHILE UNDERLINED THE SECOND

Method	San Diego I	San Diego II	HYDICE	Urban	Average
Mean	0.9935	0.9659	**1.0000**	<u>0.9973</u>	0.9892
L2-norm	0.9967	0.9959	**1.0000**	0.9954	0.9970
Max-value	0.9957	0.9740	**1.0000**	**0.9975**	0.9918
Nonlinear	0.9958	0.9938	**1.0000**	0.9969	0.9966
1×1	0.9953	0.9896	<u>0.9999</u>	0.9970	0.9955
3×3	<u>0.9984</u>	<u>0.9971</u>	0.9993	0.9971	<u>0.9980</u>
MSM	**0.9987**	**0.9979**	0.9998	<u>0.9973</u>	**0.9984**

Furthermore, the fact that neither the constrained loss nor the energy-minimized loss alone can achieve satisfactory performance highlights the inseparability inherent in the proposed CEM loss. The main reason is that, only by minimizing the background energy and maximizing the target response simultaneously, the false alarm rate can be suppressed and the detection rate increase, which is consistent with the idea of the original CEM algorithm. Sparse loss can help the deep detector further improve its performance because sparse representation aims to represent each pixel with as less the most relevant bases as possible, which is beneficial for background and target recognition. Overall, the proposed composite loss is effective, each of which has an important function and is essential.

3) Effectiveness of MSM: To investigate the effectiveness of MSM, we compare it with some other mapping operations, including the mean of each pixel (mean), the L2-norm of each pixel (L2-norm), the maximum value of each pixel (max-value), the nonlinear mapping using the Gaussian kernel function (nonlinear), the single 1×1 convolution, and the single 3×3 convolution. The nonlinear is adopted as the same form in [55], where the variance is set to 0.2. Table VII shows the performance comparison of different mapping methods. It can be observed that the proposed MSM achieves the most robust and promising performance on the four datasets. This is because the combination of 1×1 and 3×3 convolutions can maintain a good balance between position preservation and local feature utilization of targets. MSM may be limited by insufficient target spatial information or a large variation in target scale. For example, on HYDICE and Urban, the performance of MSM is slightly weaker than the mean and max-value, respectively. However, the performance gap on $AUC_{(P_f,P_d)}$ is only about 0.0002, which is within an acceptable range. The averaged results show that, by maintaining a balance of position preservation and spatial information utilization, MSM is more applicable to different datasets than pixel- and neighborhood-based methods alone, resulting in promising generalization.

VI. CONCLUSION

In this article, we proposed an IRN for HTD. To make the nonlinear representation and physical interpretability compatible, we converted a physical model into a deep learning network where the HSI can be adaptively unmixed into subspaces and coefficients of background and targets. Besides, we proposed a lightweight multiscale Transformer module to strengthen nonlinear feature representation and better extract coefficients. Moreover, we proposed a CEM loss to accurately depict the separation of background and targets. Experiments were conducted on four benchmark datasets to verify the effectiveness. The quantitative and qualitative results, including visualized detection map, ROC curves, target–background separation diagram, and AUCs, demonstrated that the proposed approach is a promising solution for HTD, and it outperforms the other competing methods. The ablation studies further verified the effectiveness of three significant elements.

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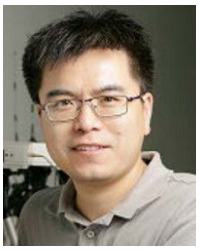
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